

The Necessity of Induction

Within Philosophy of Science, philosophers have come to find the use of induction in science to be problematic. According to Karl Popper, the problem of induction is this:

“People who say of a universal statement that we know its truth from experience usually mean that the truth of this universal can somehow be reduced to the truth of singular ones, and that those singular ones are known by experience to be true; which amounts to saying that the universal statement is based on inductive inference”. (2013, 407)

The problem that philosophers have with this is that it implies that there is an inability to be certain in any attempt to understand universal truths or laws. This essay will look at induction in terms of noumena and phenomena, as well as argue that the problem of induction is one that is inevitable; it cannot, and need not be ‘solved’.

The comparison of the ‘noumenal’ to the ‘phenomenal’ began with the philosopher Immanuel Kant. He describes the relationship as follows:

“If we entitle certain objects, as appearances, sensible entities (phenomena), then since we thus distinguish the mode in which we intuit them from the nature that belongs to them in themselves, it is implied in the distinction that we place the latter, considered in their own nature, although we do not so intuit them, or that we place other possible things, which are not objects of our senses but are thought as objects merely through the understanding, in opposition to the former, and that in doing so we entitle them intelligible entities”. (1958, 153)

Kant, then, is saying that the phenomena of something is that which we observe of that thing, while the noumena of something is the thing as it is in itself, or (as Aristotle would proclaim) the essence of said thing. Scientists, in their desire to discover universal laws, attempt to discover the noumena of the empirical world to understand these universal laws; after which, they seek to use their knowledge of the noumenal to be able to predict the future phenomena of their object of study. This, however, is not an ability that scientists possess. Scientists are only plausibly able to observe a thing's phenomenon in a series of experiments, and apply their data to what they believe to be said thing's noumena. While the former process, the process consisting of deduction, seems favorable, the latter, the inductive process, is the only one which we are able to achieve.

We are never able to be certain of a thing's noumena because to know for certain the qualities of a thing in itself, we must be able to observe a thing as it would be by itself. Anytime a scientist (or non-scientist, I suppose) observes a phenomenon, whether within nature or a controlled environment, they are studying something as it relates to other things. If we are discussing astrology, we cannot provide the location of a star without looking for it in relation to things such as the earth, or the celestial equator. We cannot study anything in chemistry as it is by itself, but only as it reacts to other chemicals, or materials. Science seeks to study the empirical world, unfortunately, the empirical world is all that we have access to. Because of this, we are only ever able to observe things in relation to other things; we are unable to entirely isolate something to learn how it is in itself, but only how it is in relation to other things.

This inability to isolate a thing cannot simply be solved by removing all the items that it may interact with; if we are able to take away everything that we think that it interacts with, we

are still met with the problem of both air particles, and air pressure. We may not consider this to be something that things can actively be in relation with, but if we place something in a different layer of the atmosphere, or in the bottom of an ocean, it will most certainly have different reactions than if it was at a regulated air pressure that can be found in a lab. Without the ability to create absolute isolation, we also lack the ability to know a thing as it is in itself. We are unable to be certain of a thing's noumena, and, as such, we are never able to decisively establish what the universal laws regarding that thing may be; despite several valiant attempts by a variety of philosophers, we are not able to eradicate the necessity of induction from scientific study.

Vacuum - perfect vacuum impossible, wouldn't work anyway bc the empirical world is dependant on operating along with other elements of the empirical world so to operate in a perfect vacuum would not be to operate as it is naturally intended to operate. (I. e. living objects in a vacuum, chemistry dependant on chemical interactions between chemicals (see previous chemistry point) how would we decide which elements are meant to be left out of the vacuum, and which ones are essential for the study/experiment to determine it's true noumena?

If any attempt to solve induction ends in vain, what then, is the alternative? I would posit that in order to understand how to cope with this scientific blow, we must look to Thomas Kuhn, and his theories about paradigm shifts (1970) to better understand what implications come with the acceptance of induction. Kuhn proposes that, as science progresses, scientists will come to new conclusions about things which they had previously believed they understood. When this happens, a paradigm shift will occur, and a new theory is developed that necessitates an

adjustment of that which scientists may have thought they knew based upon the previous paradigm.

This proposition seems to be the proper way to begin to look at induction. Scientists induce theories based upon the limited amount of phenomenon that they have observed. A continuation of their observance leads to an increase in the amount of phenomenon that they have been able to record of that thing. Theoretically, if scientists were able to observe a thing as it relates to everything; if they observed all its phenomena (which is not possible in reality because there is an infinite number of things which any given object might possibly relate to), they would be able to deduce that which makes up the thing in itself (its noumena). If we see how it relates to all possible objects, we are able to note the patterns in its behavior, that which is constant, and that which changes depending on the item or subject which it is in relation with. If this is so, then any observation that a scientist makes is necessarily allowing the scientist to come closer to discovering that thing's noumena (though, of course, this does not ensure that the conclusions drawn based upon the observation will necessarily be closer than any previous conclusions). As they continue to observe, and presumably come closer to having the ability to know a thing's noumena, they are likely to develop new theories. Though they may be getting closer to a thing's noumena, they will not and cannot come to know a thing's noumena. Because of this, the theories that they develop may be wrong.

This is where we must begin to draw from Kuhn for his theory of the paradigm shift, as well as Imre Lakatos for his theory of the 'research programme' (2013). Once new phenomena are discovered, phenomena that result in a necessary change in the current theory or law which it is believed to act under, scientists must realize or establish a new theory. While a shift is not

necessarily imperative upon discovery of a single anomaly, once a series of phenomena, or a recurring pattern of anomalies arise, a paradigm shift may be in order. A paradigm shift does not simply mean an adjustment to an already existent theory, but instead indicates a new theory, one that is able to describe the revelations that arise through newly observed phenomena. This can be observed through the shift from Newton's theory to Einstein's theory of gravitation. Newton's theory was, and still is, used to describe the observed phenomena surrounding gravity. Newton's theory, however, was found to be unable to describe higher velocities. Once the observed phenomena revealed the problem in the theory, Einstein was able to provide a new theory, which led to a paradigm shift, that accounted for the phenomena which Newton's theory could not explain.

Once the paradigm shift is made, research programmes are necessary to work through, and make corrections to, the new theory or paradigm. Evolution presents a fair image of an adjustment to the 'programme' that encompasses a theory, rather than a complete shift in paradigm. Before Darwin presented his theory of evolution, there were a few different theories about biological differences among animals. In general, it was believed that all animals were able to be categorized into certain species which were fixed and unrelated. Darwin set into motion a paradigm shift which led to the theory of evolution. Evolution as Darwin presented it, is no longer believed to be correct; however, the theory is still held to be true. Darwin sparked a paradigm shift which led to a new theory; once that theory began to be accepted, it became a research programme. Scientists have since made changes to the hypotheses that make up this theory, but the theory itself remains. Research programmes allow scientists to work within a paradigm, maintaining and editing theories to incorporate any newly observed phenomenon until

a series of phenomena has been observed that cannot be described or explained by any existing theory. After which, a paradigm shift is necessary, and a new research programme must begin within the new paradigm.

As previously mentioned, while scientists are able to observe new phenomena that allows them to come closer to knowing a thing's noumena, that does not mean that the theory that they establish will necessarily accurately describe said noumena. With every shift in paradigm comes the possibility of a shift further away from truth as well. While this is by no means reassuring, research programmes are able to provide means by which we can begin to analyze whether this new paradigm can accurately be described as a progression toward knowledge of a thing's noumena, or if it is regression presented under the guise of new phenomena. Research programmes not only serve to change a theory as needed, but also to test and analyze whether or not the new theory is valid. If, after a paradigm shift, a programme maintains a status of plausibility (while I'm sure a stronger word is desired when speaking of science, certainty cannot be promised, so we must instead work within the realm of plausibility), the programme can continue to be researched and further developed. If, however, that theory cannot be changed in any way in order to allow it to maintain its status as plausible, then scientist not only know that this paradigm is not correct, but they are now aware that the previous paradigm will no longer serve to describe phenomena resulting in a momentary lapse in any paradigm with which to work. This seemingly negative notion, the notion of being stuck without a paradigm, serves only to allow scientists to work toward discovery of a more accurate paradigm.

This process can work quicker, limiting the amount of time that scientists must attempt to function without a paradigm in a particular area (note that a paradigm shift in one area of science

does not necessitate a shift in all areas, or even multiple, though sometimes a shift does affect more than one area of scientific study) if we are aware that a shift is in fact in progress. An acknowledgement of the process of a paradigm shift is a result of the awareness that science is not certain. If we are warier of what is considered to be scientific fact, it seems that we will also be more open to new paradigms. Of course, we should not simply accept any new theory that is proposed simply because we know that science is uncertain; however, we should maintain the awareness that there are likely going to be theories developed that will come to replace the paradigm that we currently operate under. If we are more willing to shift once we run into a series of problematic phenomena, programmes are able to more rapidly either become adjusted within the current paradigmatic theory, or through their incapability indicate that a new paradigm is necessary.

If this is to progress as it should, a proper balance must be held between the paradigm shift and the research programme. If we are too quick to shift to a new paradigm, we may find ourselves in a situation where we must revisit the old paradigm and simply adjust the programme surrounding the previously held theory. If we stray too far in the other direction, we risk endlessly attempting to resuscitate a theory than should instead simply be replaced. Scientists working within a programme should consistently seek to edit and correct the current paradigm, yet also be willing to test a new theory if they run into an abundance of unexplained phenomena within the current programme.

When we come to accept this process, we avoid prolonging the inevitable shift. In the past, we have seen a hesitance to accept a new theory due to the belief that science is certain, theories have been rejected for years, only to later resurface and produce the inevitable paradigm

shift. The most prominent example of this is in the shift from geocentric to heliocentric models of the universe. The geocentric model had been believed for many centuries due largely to religious beliefs. There had been speculation about the possibility of the earth moving, however, a heliocentric model had not been entirely developed until the 16th century with Copernicus. Because of the certainty that people believed this science to produce, they did not believe the Copernican model, and it was still believed to be heretical a century later when Galileo strongly advocated for heliocentricity. Even with Galileo, people were hesitant to accept a change in their scientific beliefs, and it took a lot more persuasion before heliocentricity was even considered among the masses.

This may appear to be an extreme and outdated example; however, it merely serves to display the danger of believing science to be entirely correct. If we credit science with the same amount of certainty as those in the 16th century, then we merely begin to delay the shift that, with the collection of more observed phenomena, becomes necessary. If, however, we accept that science cannot be certain, and that paradigm shifts must occur when current theories become evidently insufficient, we are able to more quickly progress to a further understanding of the phenomena that makes up a thing's noumena.

Current philosophers of science seek to find a way to eradicate the problem of induction. Because of our limited scope of observable phenomena, this is not, in reality, possible. Rather than turning induction into a problem that must be solved, we should instead allow that induction cannot be solved, and therefore should not be viewed as problematic, but inevitable. If we are to shift away from the view of induction as a problem, we must make a shift toward something else. My proposition, then, is to move toward a balance between research programmes that are able to

edit and maintain a theory until it is no longer able to explain the phenomena that has been observed (ideally it would explain all phenomena, but as was explained above, if we were able to know all phenomena, we would have the ability to know a thing's noumena, and we would not need theories for we would have certainty), and paradigm shifts that are able to explain the new observations. With an acknowledgement of this process comes the ability to know when it is happening and to avoid hindering or prolonging this process as has been done in the past.

References

Cover, J. A., Martin Curd, and Christopher Pincock, eds. 2013. *Philosophy of Science: The Central Issue*. 2nd Edition. New York: W. W. Norton & Company.

Kant, Immanuel. 1958. *Critique of Pure Reason*. Trans. Norman Kemp Smith. New York: Modern Library.

Kuhn, Thomas. 1970. *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press.

Lakatos, Imre. 2013. "Science and Pseudoscience". In Cover et al. 2013, 20-26.

Popper, Karl. 2013. "The Problem of Induction". In Cover et al. 2013, 406-411.